

MAGIC VALLEY, ID, USA

WMO: 725866

Lat: 42.482N

Lon: 114.487W

Elev: 1265

StdP: 87.02

Time Zone: -7.00 (NAM)

Period: 97-19

WBAN: 94178

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-13.9	-11.3	-18.1	0.9	-12.1	-15.5	1.1	-9.0	14.9	2.5	13.2	1.6	3.0	100	0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	35.0	17.0	33.4	16.7	32.1	16.3	19.0	31.4	18.0	30.1	17.2	29.2	3.9	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	12.1	23.6	13.0	10.9	23.5	12.3	10.4	23.2	59.0	31.5	55.7	29.9	52.9	29.2	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	11.0	9.3	DB	-17.7	37.9	3.7	1.2	-20.4	38.8	-22.5	39.5	-24.6	40.2	-27.2	41.1	(4)
(5)			WB	-18.1	20.8	3.6	1.2	-20.7	21.7	-22.8	22.4	-24.8	23.0	-27.5	23.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	-1.0	1.1	5.6	8.6	13.6	18.5	24.0	22.7	17.5	10.3	3.6	-1.4	(6)	
	DBStd	9.65	4.83	4.40	4.05	4.09	4.56	4.26	3.06	3.08	4.30	4.67	4.98	5.11	(7)	
	HDD10.0	1434	341	249	146	73	18	1	0	0	3	53	197	353	(8)	
	HDD18.3	3353	599	483	394	291	159	49	3	6	65	252	441	611	(9)	
	CDD10.0	1552	0	1	9	33	128	257	434	394	229	62	6	0	(10)	
	CDD18.3	429	0	0	0	1	12	56	178	141	41	2	0	0	(11)	
	CDH23.3	5414	0	0	1	19	192	770	2186	1664	541	42	0	0	(12)	
	CDH26.7	2427	0	0	0	2	51	302	1109	773	181	8	0	0	(13)	
(14) Wind	WSAvg	4.6	4.3	4.9	5.2	5.5	4.9	4.8	4.3	4.2	4.3	4.6	4.4	4.4	(14)	
(15) Precipitation	PrecAvg	272	27	18	24	32	38	21	9	13	16	22	23	29	(15)	
	PrecMax	434	61	43	48	100	105	117	21	134	64	70	52	81	(16)	
	PrecMin	146	7	6	8	5	8	0	1	0	0	2	8	0	(17)	
	PrecStd	82	14	10	13	21	27	24	6	28	14	18	13	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	14.7	22.1	25.9	31.0	34.6	37.5	36.2	33.0	27.5	19.1	13.3	(19)	
		MCWB	6.6	7.0	9.6	12.2	15.0	17.2	18.7	17.5	16.3	13.0	9.4	6.7	(20)	
	2%	DB	8.8	11.7	18.3	22.6	27.8	32.3	35.8	34.1	30.9	23.8	16.2	10.1	(21)	
		MCWB	4.7	6.0	8.5	10.7	14.3	16.1	17.2	16.8	15.5	11.9	8.2	5.9	(22)	
	5%	DB	7.1	9.2	15.9	19.7	25.4	30.1	34.1	32.8	28.7	21.2	13.8	7.7	(23)	
		MCWB	3.7	4.8	7.4	9.8	13.0	15.4	16.9	16.5	14.6	10.8	7.2	4.2	(24)	
	10%	DB	5.3	7.5	13.0	17.0	22.6	27.9	32.6	31.3	26.7	18.7	11.4	5.9	(25)	
		MCWB	2.4	3.9	6.3	8.6	12.2	14.7	16.6	16.2	13.8	9.6	6.0	2.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	7.9	10.6	13.4	16.9	19.0	20.6	19.9	18.7	14.1	10.7	7.6	(27)	
		MCDB	10.6	13.0	18.4	23.5	27.6	32.2	33.2	32.9	28.2	24.6	17.0	11.2	(28)	
	2%	WB	5.3	6.6	9.2	11.5	15.2	17.5	19.2	18.6	16.9	12.6	9.0	6.1	(29)	
		MCDB	8.3	10.8	16.8	20.4	25.1	29.6	31.8	31.0	27.4	21.8	15.0	10.0	(30)	
	5%	WB	3.9	5.4	8.0	10.2	13.9	16.2	18.3	17.7	15.6	11.4	7.7	4.6	(31)	
		MCDB	6.6	8.9	14.9	18.4	23.4	27.7	30.6	29.5	26.1	19.5	13.0	7.5	(32)	
	10%	WB	2.8	4.1	6.8	9.0	12.7	15.2	17.5	16.8	14.5	10.2	6.4	3.0	(33)	
		MCDB	5.0	7.2	12.5	16.2	21.6	26.2	29.7	28.7	25.1	17.8	10.9	5.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.6	11.4	12.3	13.3	14.7	15.8	15.6	14.9	13.0	11.0	8.2	(35)	
		MCDBR	9.6	10.8	15.6	17.0	17.1	18.0	17.6	17.9	18.0	16.5	14.3	9.5	(36)	
	5% WB	MCWBR	6.5	6.9	8.3	8.4	7.7	8.0	7.3	7.7	8.3	8.3	8.0	6.2	(37)	
		MCWBR	8.9	10.1	14.2	15.3	15.7	17.2	16.7	16.7	16.1	14.7	13.0	9.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.250	0.260	0.273	0.295	0.315	0.311	0.325	0.323	0.297	0.276	0.258	0.250	(40)	
		taud	2.573	2.589	2.572	2.501	2.473	2.525	2.495	2.478	2.547	2.590	2.595	2.576	(41)	
	Ebn at Noon		924	967	985	976	958	959	941	934	942	931	906	894	(42)	
		Edh at Noon	65	76	87	102	108	103	105	103	89	75	63	60	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.79	2.63	4.02	5.49	6.62	7.65	7.53	6.55	5.27	3.58	2.18	1.43	(44)	
	RadStd		0.21	0.33	0.34	0.38	0.58	0.52	0.41	0.33	0.31	0.24	0.19	0.21	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-0.71	-0.89	N/A	N/A	N/A	N/A	(46)
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page